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$$\begin{aligned} & \int \frac{dx}{\cos^4(x)} \\ &= \int \sec^4(x) dx \\ &= \int \sec^2(x)(\tan^2(x) + 1) dx \\ & t = \tan(x), \quad dt = \sec^2(x) dx \\ &= \int (t^2 + 1) dt \\ &= \int t^2 dt + \int dt \\ &= \frac{t^3}{3} + t + C \\ &= \frac{1}{3} \tan^3(x) + \tan(x) + C \end{aligned}$$

References